5.5 Differentiable Functions

Definition 5.5.1 (Differentiable Scalar Function)
*Let K be a complete valued field, $U \subset K$ an open subset, Y a normed vector space over K and $f : U \to Y$ a function. We say that f is **differentiable** at $x \in U$ if the following relation holds*

$$\forall \varepsilon > 0 \exists \delta > 0 \text{ s.t. } 0 < |y - x| < \delta \Longrightarrow \left\| \frac{f(y) - f(x)}{y - x} - f'(x) \right\| < \varepsilon$$

for some $f'(x) \in K$.

*We say that f is simply **differentiable** if it is differentiable at every point of U.*

Trivially if f is differentiable then so is $f|_V$ for any open subset $V \subset U$.

Proposition 5.5.2
Let K be a valued field, $U \subset K$ an open subset and $f : U \to Y$ a function which is differentiable at $x \in U$. Suppose $x_n \to x$ is a sequence such that $x_n \neq x$. Then

$$\frac{f(x_n) - f(x)}{x_n - x} \to f'(x)$$

Proposition 5.5.3
Let K be a valued field, $U \subset K$ an open subset and $f : U \to Y$ a function which is differentiable at $x \in U$. Then f is continuous at x.

Proposition 5.5.4 (Arithmetic of Differentiable Functions)
Let K be a valued field, $U \subset K$ an open subset and $f, g : U \to Y$ be functions differentiable at $x \in U$. Then

a) $(f + g)'(x) = f'(x) + g'(x)$
b) $(\lambda f)'(x) = \lambda f'(x)$

If Y has continuous algebra structure (e.g. $Y = K$) then

$$(fg)'(x) = f'(x)g(x) + f(x)g'(x)$$

and if $Y = K$ and f is non-zero on U then

$$\left(\frac{1}{f}\right)'(x) = -f' \frac{x}{f(x)^2}$$

Finally every linear (and therefore constant) function is trivially differentiable.

Proposition 5.5.5
Let K be a valued field, $U \subset K$ an open subset and $U = \bigcup_{\alpha \in \mathcal{A}} V_\alpha$ an open cover. Suppose $f : U \to Y$ is a function. Then f is (continuously) differentiable if and only if $f|_{V_\alpha}$ is (continuously) differentiable for all $\alpha \in \mathcal{A}$.

Corollary 5.5.6
Let K be a valued field of characteristic zero and n a positive integer. Then the derivative of x^n is nx^{n-1}.